

JANGARA BLISS

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EDUCATION

Fort Lewis College — B.S. Computer Engineering

May 2027

Minors: Mathematics & Business Administration · GPA 3.65 · Upper-division GPA 3.8

RESEARCH EXPERIENCE

Research Assistant — Humanoid Robotics

May 2026 – Present

Fort Lewis College · Advisor: Dr. Yiyan Li · Independent undergraduate research

- Integrated the Booster K1 into VLN-CE-Isaac / NaVILA and completed all **1,077** R2R val-unseen simulation episodes: **18.3% success, 30.3% oracle success**, 10.9% SPL, and 7.59 m final navigation error.
- Decomposed outcomes into 751 never-reached, 129 reached-without-stopping, and 197 successful episodes, separating exploration from arrival recognition and motivating a controlled viewpoint study.
- Extended upstream K1 training/deployment surfaces with environment configuration, reward and observation-contract checks, wrappers, validators, command filtering, and safety-aware mode handling; canonical results include per-episode artifacts and a public evidence ledger.

Independent Research — xembench

2026

Language-grounded manipulation across a Franka arm and Unitree G1 in ManiSkill3

- Designed and executed a two-embodiment benchmark with a canonical 21-D action interface, 1,202 demonstrations, 18 BC policies, and a **73-cell / 6,550-episode** baseline matrix; reported near-zero transfer under weak native precision policies.
- Ran a failure-driven data round and intervention study; action chunking improved tool-pulling success from **1.3% to 14.7%** across three seeds, while grasp-critical tasks remained near zero. AI-assisted implementation is documented in the repository.

Research Assistant — NASA Colorado Robotics Challenge

Oct 2025 – Apr 2026

Fort Lewis College · Four-person team

- Led integration for an 18-DoF hexapod fielded at Great Sand Dunes; worked on Arduino/IMU/bump-sensor/power integration and a three-bit behavior interface, with team and inherited-code provenance stated publicly.
- Built a later Python Kuramoto coupled-oscillator gait simulator and 34-test CI suite; the simulator is presented separately from the physical challenge firmware.

Research Assistant — AI & Robotics

May – Sep 2025

Fort Lewis College · Advisor: Dr. Kevin Wedeward · Solo project

- Restored legacy Sawyer and Baxter platforms, rebuilt a ROS Noetic / MoveIt / Gazebo workstation, and developed Blender/Python synthetic-data and YOLO inspection tooling; no unsupported detector metric or live closed-loop result is claimed.

SELECTED TECHNICAL WORK

KUKA KR 6 Kinematics & Verification

2026

Individual robotics course project; later packaged as a public library

- Implemented FK, closed-form position IK, a 6×6 geometric Jacobian, singularity diagnostics, inverse velocity kinematics, and trajectories; verified with finite differences, 200 random round trips, a generated URDF, and **39 tests**.

CLiDA LiDAR Debris Analysis

Fall 2024

Algorithm developer on a five-person design team

- Implemented point-cloud pruning/conversion, dimension, PCA-axis, motion, and ICP rotation-rate methods; final report records 1.009 and 3.004 RPM means with all 20 trials within ±0.2 RPM.

TECHNICAL SKILLS

Languages: Python, C++, Verilog, JavaScript / TypeScript, SQL, Bash

Robot learning / AI: PyTorch, behavior cloning, PPO, NaVILA, vision-language navigation, YOLO, experiment design

Robotics: Isaac Sim / Lab, ManiSkill3, MuJoCo, ROS / ROS 2, MoveIt, Gazebo, kinematics, motion planning

Hardware / tools: Arduino, FPGA / Vivado, PCB design, sensors, Git, Docker, Linux, AWS, PostgreSQL, Neo4j

LEADERSHIP & HONORS

Student Representative, FLC Strategic Implementation Committee · President & Co-Founder, Entrepreneurial Ventures Association (8-person executive team; \$1,500 in student micro-grants) · **Physics & Engineering Symposium 2nd Place** (2025) · **Dean's List** (Spring 2026) · **Goldman Sachs Emerging Leaders** (2024)